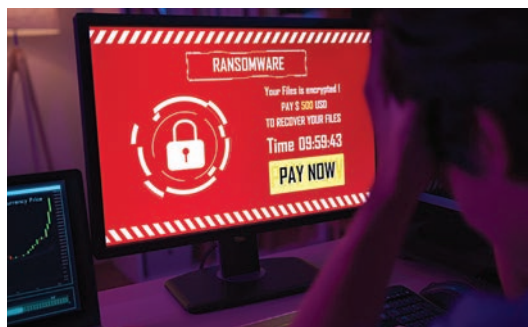


group (APT29). The attackers compromised SolarWinds' Orion software, a widely used IT management tool, by inserting malicious code into legitimate updates. Over 18,000 organisations, including Fortune 500 companies, U.S. government agencies, and nuclear laboratories, installed the tainted updates, which deployed the SUNBURST malware. This created backdoors, allowing



hackers to exfiltrate sensitive data over months, undetected. Targets included the Pentagon, the Department of Homeland Security, and Microsoft, with an estimated 50 million records exposed.

The attack's sophistication lay in its stealth and scale. By exploiting a trusted software provider, the hackers bypassed traditional defenses, highlighting the fragility of supply chains. The breach exploited a zero-day vulnerability and weak authentication protocols, enabling persistent access to high-value networks. The U.S. government responded with sanctions against Russia and new cybersecurity mandates, including mandatory breach disclosures and enhanced vendor scrutiny. Globally, organisations adopted software bill of materials (SBOM) frameworks to ensure supply chain integrity. The SolarWinds attack remains a cautionary tale of how third-party vulnerabilities can cascade into catastrophic breaches, urging a shift toward zero-trust architectures and rigorous software auditing.

3. LockBit ransomware group

The LockBit ransomware group, emerging in the early 2020s, has become a dominant force in cyber-crime, operating as a ransomware-as-

a-service (RaaS) syndicate. LockBit targets organisations worldwide, including India's Motilal Oswal Financial Services and Polycab India, encrypting systems and demanding cryptocurrency ransoms. Their double-extortion tactics involve stealing sensitive data before encryption and threatening to leak it if payments are not made. LockBit exploits unpatched vulnerabilities, such as SQL injection flaws, and uses sophisticated phishing campaigns to gain initial access. In 2023, a reported counter-hack disrupted LockBit's operations, leaking internal communications, but the group quickly regrouped, showcasing its resilience.

LockBit's impact is staggering, with attacks disrupting critical sectors like finance, healthcare, and manufacturing. In India, their targeting of financial firms exposed customer data, eroding trust. The group's adaptability, leveraging tools like Cobalt Strike and custom ransomware variants, makes it a persistent threat. Law enforcement efforts, including international operations like Operation Cronos in 2024, have disrupted LockBit's infrastructure, but its decentralised model ensures longevity. The rise of LockBit has driven companies to prioritise patch management, robust backups, and response plans, while highlighting the need for global cooperation to combat RaaS networks.

4. Transparent Tribe (APT36)

Transparent Tribe, or APT36, a Pakistan-aligned advanced persistent threat (APT) group, launched a cyberattack in May 2025, targeting India's critical infrastructure. The campaign attacked defense, healthcare, telecom, and government sectors, including AIIMS, Jio, and BSNL, using spear phishing, malicious scripts, and distributed denial-of-service (DDoS) attacks.

APT36 deployed the Ares RAT for keylogging, screen capturing, and credential theft, exploiting geopolitical events like the Pahalgam terror attack to craft convincing phishing lures. Over 650 DDoS and defacement incidents were recorded, amplified by hacktivist groups aligned with APT36's objectives.

While it didn't cause any major fallout, the campaign's hybrid approach, blending espionage with disruptive attacks, did expose few gaps in India's cybersecurity posture, particularly in network segmentation and authentication. India responded with enhanced threat intelligence sharing and continued investments in critical infrastructure protection. APT36's persistence underscores the need for nations to bolster defenses against state-backed actors, integrating real-time monitoring and international cooperation to counter hybrid threats.

5. Hathway and BSNL breaches

India's telecom sector faced breaches with Hathway and BSNL, often linked to broader attack campaigns. Hathway, a major ISP, suffered a breach in the early 2020s, exposing over 41.5 million customer records due to a vulnerability in its content management, exploited by a hacker known as "dawnofdevil."

The leaked data included names, phone numbers, and addresses, posing risks of identity theft and phishing. Similarly, BSNL, India's state-owned telecom provider, was targeted under recent India-Pakistan conflict in 2025, facing DDoS attacks and data leaks that disrupted services and compromised user information.

